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Re-NEET Answer Key 2026 Code-70

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Physics (Paper Code 70)

Q1. A photon and an electron, each having energy 20 eV, move in free space. The ratio of the linear momentum of the electron to that of the photon is:

Options:

- A. 275
- B. $450/2$
- C. $1/250$
- D. 225

Answer: D. 225

Q2. Water flows in streamline motion through a horizontal pipe. The pressure difference between points P and Q is 15 N m^{-2} . The cross-sectional areas at P and Q are 40 cm^2 and 20 cm^2 respectively. The rate of flow of water through the pipe is:

Options:

- A. $400 \text{ cm}^3 \text{ s}^{-1}$
- B. $100 \text{ cm}^3 \text{ s}^{-1}$
- C. $200 \text{ cm}^3 \text{ s}^{-1}$
- D. $300 \text{ cm}^3 \text{ s}^{-1}$

Answer: A. $400 \text{ cm}^3 \text{ s}^{-1}$



Q3. A thin horizontal disc rotates about a vertical axis through its fixed centre O. Its angular momentum about points A and B are L_A and L_B respectively, with $OB = 2OA$. The value of L_A/L_B is:

Options:

- A. 2
- B. $1/4$
- C. $1/2$
- D. 1

Answer: D. 1

Q4. Consider a long solenoid of length l and radius r . If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is:

Options:

- A. $2\mu_0\pi n^2 r^2 l$
- B. $\mu_0\pi n^2 r^2 l$
- C. $\mu_0 n^2 r^2 l$
- D. $(\mu_0\pi n^2 r^2 l)/2$

Answer: B. $\mu_0\pi n^2 r^2 l$

Q5. The temperature of a metallic sphere of radius R is increased by a small amount ΔT . If the linear coefficient of thermal expansion of the metal is α , the approximate increase in the volume of the sphere is:

Options:

- A. $6\pi R^3\alpha\Delta T$
- B. $2\pi R^3\alpha\Delta T$
- C. $3\pi R^3\alpha\Delta T$
- D. $4\pi R^3\alpha\Delta T$

Answer: D. $4\pi R^3\alpha\Delta T$

Q6. Consider two circuits (A) and (B), each having two resistors. One has a positive temperature coefficient of resistance ($+\alpha$) and the other has a negative temperature coefficient ($-\alpha$). As temperature increases, which statement is correct?

Options:

- A. Both I_A and I_B remain constant
- B. I_A remains constant while I_B increases
- C. I_A decreases while I_B increases
- D. I_A increases while I_B decreases

Answer: B. I_A remains constant while I_B increases

Q7. A beam of light falls on a metal surface such that photoelectrons are generated. If the power of the light source decreases linearly with time, then the variation of photocurrent I and stopping potential $|V|$ with time is best represented by:

Options:

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4

Answer: B. Option 2

Q8. An ideal Zener diode with breakdown voltage of -3 V is reverse biased with an input voltage of -5 V . The magnitude of the voltage difference between points B and A is:

Options:

- A. 0 V
- B. 3 V
- C. 2 V
- D. 1 V

Answer: C. 2 V

Q9. In the measurement of viscosity of liquids using the terminal velocity experiment, spherical balls of the same radius but different densities are used. The variation of terminal velocity with density ratio is best represented by:

Options:

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4

Answer: B. Option 2

Q10. Two planets P_1 and P_2 have equal masses and radii R_1 and R_2 respectively, where $R_1/R_2 = 2$. If their escape speeds are v_1 and v_2 , then v_2/v_1 is:

Options:

- A. 2
- B. $1/2$
- C. 1
- D. $\sqrt{2}$

Answer: D. $\sqrt{2}$

Q11. An AC voltage $V = 220 \sin(2 \times 10^3 t)$ volt is applied to a series LCR circuit. Given $L = 10 \text{ mH}$, $C = 25 \text{ } \mu\text{F}$ and $R = 100 \text{ } \Omega$. The current amplitude is:

Options:

- A. 22.0 A
- B. 2.2 A
- C. 5.5 A
- D. 11.0 A

Answer: B. 2.2 A



Q12. Two identical inductors are connected in configurations P and Q. If the induced emfs are E_P and E_Q respectively, then E_P/E_Q is:

Options:

- A. 2
- B. $1/4$
- C. $1/2$
- D. 1

Answer: A. 2

Q13. Three identical capacitors P, Q and S each having capacitance C are connected to a battery of voltage V . If the energy stored in capacitor P is U_P and total energy stored is U_T , then U_P/U_T is:

Options:

- A. $1/6$
- B. $2/3$
- C. $1/3$
- D. $1/2$

Answer: A. $1/6$

Q14. A conducting loop of finite resistance lies in the x-y plane. Its area varies with time as $A = A_0(1 + \sin\omega t)$ in a constant magnetic field. The qualitative variation of power dissipated with time is represented by:

Options:

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4

Answer: C. Option 3

Q15. In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas decreases from 60 K to 50 K. The work done by the gas is:

Options:

- A. 166 J
- B. 41.5 J
- C. 83 J
- D. 124.5 J

Answer: D. 124.5 J

Q16. Consider a particle moving along a straight line whose position is given by $s(t) = \alpha t^2 + \beta t + \gamma$, where $\alpha = -1 \text{ m s}^{-2}$, $\beta = 6 \text{ m s}^{-1}$ and $\gamma = 5 \text{ m}$. The average speed of the particle from $t = 0 \text{ s}$ to $t = 6 \text{ s}$ is:

Options:

- A. 0 m s^{-1}
- B. 12 m s^{-1}
- C. 6 m s^{-1}
- D. 3 m s^{-1}

Answer: D. 3 m s^{-1}



Q17. Match the following parts of the electromagnetic spectrum with their major applications.

Part Spectrum

- P Microwave
- Q UV Rays
- R Gamma Rays
- S Radio Waves

Application

- I. Purifying water
- II. Warming food
- III. AM/FM communication

Application

IV. Treating cancer cells

Options:

- A. P-II, Q-IV, R-III, S-I
- B. P-I, Q-II, R-III, S-IV
- C. P-I, Q-IV, R-II, S-III
- D. P-II, Q-I, R-IV, S-III

Answer: D. P-II, Q-I, R-IV, S-III

Q18. An ideal gas is made of polyatomic molecules. Each molecule has three translational, three rotational and f vibrational modes. If $C_p/C_v = 8/7$, then the value of f is:

Options:

- A. 1
- B. 4
- C. 3
- D. 2

Answer: B. 4

Q19. A unit positive point charge is taken slowly through a thin tube inside a uniformly charged dielectric sphere of radius R and charge density ρ . The magnitude of the total work done is $(\rho R^2)/(n\epsilon_0)$. The value of n is:

Options:

- A. 18
- B. 2
- C. 6
- D. 9

Answer: A. 18

Q20. A current I_0 flows through a metallic circular loop of radius r . The resistance of segment ABC is half that of segment ADC. The magnetic field at the centre O is:

Options:

- A. $\mu_0 I_0 / 2\pi r$
- B. $\mu_0 I_0 / 12r$
- C. $\mu_0 I_0 / 4r$
- D. $\mu_0 I_0 / 2r$

Answer: B. $\mu_0 I_0 / 12r$

Q21. Bob B of mass m hangs from a string of length 10 m. A point mass A of mass m moving horizontally at 10 m s^{-1} collides elastically with B. The maximum height reached by B after collision is:

Options:

- A. 2.5 m

- B. 8 m
- C. 7 m
- D. 5 m

Answer: D. 5 m

Q22. An electromagnetic wave travels in a dielectric medium with dielectric constant $\epsilon_r = 9$. Which of the following statements is incorrect?

Options:

- A. The wave propagates along +z direction
- B. Speed of wave = 10^8 m s^{-1}
- C. Wavelength = 300 m
- D. Magnetic field follows the given sinusoidal relation

Answer: C. Wavelength = 300 m

Q23. A particle of mass M moves from $x = 0$ to $x = L$. The coefficient of kinetic friction varies as $\mu_k(x) = \mu_0 - \alpha x$ and $\mu_k(L) = 0$. The total work done by friction is $n\mu_0 MgL$. Find n.

Options:

- A. 1/2
- B. 3
- C. 1
- D. 1/3

Answer: A. 1/2



Q24. Three media P, Q and R have refractive indices 1, 1.25 and 1.5 respectively. Medium Q has thickness 5 cm and the object is at its centre. If apparent depths viewed from P and R are h_1 and h_2 respectively, then $|h_1 - h_2|$ is:

Options:

- A. 3 cm
- B. 0 cm
- C. 1 cm
- D. 2 cm

Answer: C. 1 cm

Q25. A uniformly charged insulating sphere of radius R and charge +Q attracts a point charge $-q$ released from rest at distance $3R$. The speed of the charge when it reaches the sphere surface is:

Options:

- A. $\sqrt{Qq/4\pi\epsilon_0 mR}$
- B. $\sqrt{3Qq/4\pi\epsilon_0 mR}$
- C. $\sqrt{2Qq/3\pi\epsilon_0 mR}$
- D. $\sqrt{Qq/3\pi\epsilon_0 mR}$

Answer: D. $\sqrt{Qq/3\pi\epsilon_0 mR}$

Q26. A car moves on a circular banked road of radius 50 m at speed 10 m s^{-1} . If wear and tear on tyres is minimum, the banking angle θ is:

Options:

- A. $\tan^{-1}(2/3)$
- B. $\tan^{-1}(1/5)$
- C. $\tan^{-1}(2/5)$
- D. $\tan^{-1}(3/2)$

Answer: B. $\tan^{-1}(1/5)$

Q27. Two unit masses move in opposite directions on a frictionless circular wire with equal angular speeds. Which graph correctly represents the magnitude of total linear momentum versus angle θ ?

Options:

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4

Answer: D. Option 4

Q28. Three identical p-n junction diodes D_1 , D_2 and D_3 are connected as shown. If their depletion widths are W_1 , W_2 and W_3 respectively, then:

Options:

- A. $W_2 > W_1 = W_3$
- B. $W_1 > W_2 > W_3$
- C. $W_3 = W_1 > W_2$
- D. $W_3 > W_2 > W_1$

Answer: D. $W_3 > W_2 > W_1$



Q29. Two lenses of focal lengths +10 cm and -10 cm are arranged as shown. The position of the final image is:

Options:

- A. 60 cm to the right of the concave lens
- B. 20 cm to the left of the concave lens
- C. 60 cm to the left of the concave lens
- D. 30 cm to the right of the concave lens

Answer: C. 60 cm to the left of the concave lens

Q30. A solid sphere A (radius R , mass M) is attached to a smaller solid sphere B (radius r , mass m). The moments of inertia about vertical axes through A and B are I_A and I_B respectively. The value of $I_A - I_B$ is:

Options:

- A. 0
- B. $(M - m)(R + r)^2$
- C. $(m - M)(R + r)^2$
- D. $(m - M)(R - r)^2$

Answer: C. $(m - M)(R + r)^2$

Physics (Paper Code 70)

Q31. Two identical charged particles are placed at a certain distance apart. If the force between them is F and the distance is doubled while the charges remain unchanged, the new force becomes:

Options:

- A. F
- B. $F/2$
- C. $F/4$
- D. $4F$

Answer: C. $F/4$

Q32. Two gases A and B are kept at the same temperature. If the root mean square speed of gas A is twice that of gas B, then the relation between the molecular masses is:

Options:

- A. $n_A = 2n_B$
- B. $n_A = n_B/2$
- C. $n_A = 4n_B$
- D. $n_A = n_B$

Answer: B. $n_A = n_B/2$

Q33. Four identical resistors are connected in a network. The equivalent resistance between the given terminals is:

Options:

- A. R
- B. $R/2$
- C. $R/4$
- D. $4R$

Answer: C. $R/4$

Q34. A body projected vertically upward returns to the point of projection. Neglecting air resistance, the ratio of its initial kinetic energy to final kinetic energy is:

Options:

- A. 1
- B. 0
- C. 2
- D. $1/2$

Answer: A. 1

Q35. The energy released in the fusion reaction involving deuterium and tritium is closest to:

Options:

- A. 17.6 MeV



- B. 3.726 MeV
- C. 3726 keV
- D. 1.76 MeV

Answer: C. 3726 keV

Q36. In a laboratory experiment, the focal length of a convex lens is determined using:

Options:

- A. Screw gauge
- B. Vernier calipers
- C. Optical bench
- D. Sonometer

Answer: C. Optical bench

Q37. A planet revolves around a star in a circular orbit. If the orbital radius becomes R , the time period varies as:

Options:

- A. R
- B. R^2
- C. $R^{3/2}$
- D. $\sqrt{R}$

Answer: C. $R^{3/2}$



Q38. The dimensional formula of a physical quantity is:

Options:

- A. $[LT^{-1}K^{-2}]$
- B. $[MLT^{-2}]$
- C. $[M^0LT^{-1}]$
- D. $[ML^2T^{-2}]$

Answer: A. $[LT^{-1}K^{-2}]$

Q39. Three points V_1 , V_2 and V_3 are situated in an electric field. The correct relation among their electric potentials is:

Options:

- A. $V_1 = V_2 = V_3$
- B. $V_1 > V_2 > V_3$
- C. $V_1 = 0, V_2 < V_3$
- D. $V_1 < V_2 = V_3$

Answer: C. $V_1 = 0, V_2 < V_3$

Q40. A convex mirror forms an image of an object. The distance between the object and image is measured. The focal length of the mirror is:

Options:

- A. 2.08 cm
- B. 1.04 cm
- C. 4.16 cm
- D. 0.52 cm

Answer: B. 1.04 cm

Q41. Three points A, B and C are located in an electric field due to a charge distribution. The magnitudes of electric fields at these points satisfy:

Options:

- A. $E_A = E_B = E_C$
- B. $E_A \neq 0, E_B = E_C$
- C. $E_A = 0, E_B \neq E_C$
- D. $E_A > E_B > E_C$

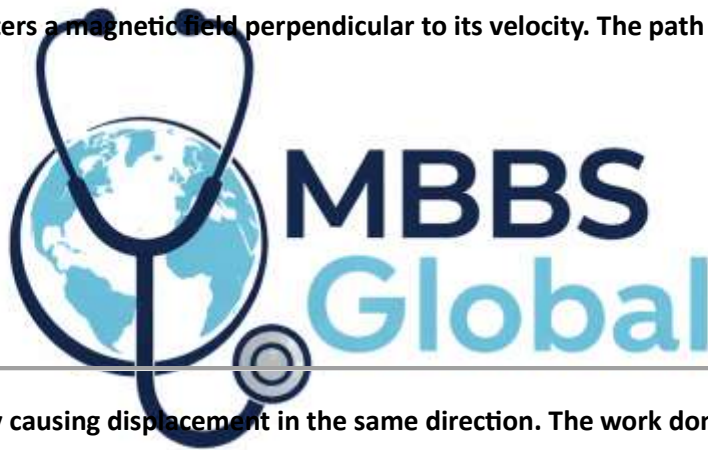
Answer: B. $E_A \neq 0, E_B = E_C$

Q42. A charged particle enters a magnetic field perpendicular to its velocity. The path followed by the particle is:

Options:

- A. Straight line
- B. Parabola
- C. Ellipse
- D. Circle

Answer: D. Circle



Q43. A force acts on a body causing displacement in the same direction. The work done is:

Options:

- A. 300 J
- B. 600 J
- C. 900 J
- D. 1200 J

Answer: B. 600 J

Q44. Two long parallel conductors carrying equal currents attract each other. The force per unit length between them is:

Options:

- A. $\mu_0 I^2 / (\pi \lambda g)$
- B. $\mu_0 I^2 / 2\pi d$
- C. $\mu_0 I^2 / 4\pi d$
- D. $\mu_0 I^2 / \pi d$

Answer: A. $\mu_0 I^2 / (\pi \lambda g)$

Q45. In a moving coil galvanometer, the figure of merit (k_m) is found to be:

Options:

- A. 0.5
- B. 1
- C. 2
- D. 4

Answer: B. 1

End of Physics (Paper Code 70)

Chemistry (Paper Code 70)

Q46. Consider the following reaction and choose the correct option.

Aromatic compound $\rightarrow$ (i) $\text{CrO}_2\text{Cl}_2 / \text{CS}_2$ (ii) $\text{H}_3\text{O}^+ \rightarrow \text{P}$

Options:

- A. Compound P is obtained by the hydrogenation of benzoyl chloride with Pd on BaSO_4 .
- B. On treating compound P with saturated NaHCO_3 solution, brisk effervescence is observed.
- C. Compound P can be prepared by treating benzene with anhydrous AlCl_3 and CH_3COCl .
- D. On treatment with bromine water, compound P gives a white precipitate.

Answer: A. Compound P is obtained by the hydrogenation of benzoyl chloride with Pd on BaSO_4 .

Q47. The formula of tetraammineaquachloridocobalt (III) chloride is:

Options:

- A. $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$
- B. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2(\text{H}_2\text{O})]$
- C. $[\text{Co}(\text{NH}_3)_3\text{Cl}_4(\text{H}_2\text{O})]$
- D. $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})]\text{Cl}_3$

Answer: A. $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$

Q48. The lanthanide ion having four unpaired electrons is:

(Given: Ce = 58, Nd = 60, Tb = 65 and Ho = 67)

Options:

- A. Ho^{3+}

- B. Nd^{3+}
- C. Ce^{3+}
- D. Tb^{3+}

Answer: A. Ho^{3+}

Q49. For an elementary chemical reaction, the Arrhenius plot is given. If the activation energy is 6.64 kJ mol^{-1} and $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$, the temperature at which the rate constant becomes $\text{e}^{21} \text{ min}^{-1}$ is:

Options:

- A. 250 K
- B. 125 K
- C. 150 K
- D. 200 K

Answer: D. 200 K

Q50. The green paramagnetic species formed by heating KMnO_4 at 513 K is:

Options:

- A. KO_2
- B. K_2MnO_4
- C. Mn_3O_4
- D. MnO

Answer: B. K_2MnO_4



Q51. Given below are two statements:

Statement I: trans-But-2-ene upon treatment with Br_2 in CCl_4 gives the shown product.

Statement II: cis-But-2-ene upon treatment with alkaline KMnO_4 gives the shown product.

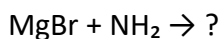
Choose the correct answer.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: A. Statement I is incorrect but Statement II is correct

Q52. One of the products formed in the following reaction is:



Options:

- A. Structure (1)

- B. Structure (2)
- C. Structure (3)
- D. Structure (4)

Answer: A. Structure (1)

Q53. Given below are two statements:

Statement I: Heating NaCl with concentrated H_2SO_4 and MnO_2 results in oxidation of Mn.

Statement II: Heating NaI with concentrated H_2SO_4 and MnO_2 results in reduction of Mn.

Choose the correct answer.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: A. Statement I is incorrect but Statement II is correct

Q54. Among the following options, the correct trend in electron gain enthalpy is:

Options:

- A. $\text{I} > \text{Br} > \text{Cl} > \text{F}$
- B. $\text{F} > \text{Cl} > \text{Br} > \text{I}$
- C. $\text{Br} > \text{Cl} > \text{F} > \text{I}$
- D. $\text{Cl} > \text{F} > \text{Br} > \text{I}$

Answer: D. $\text{Cl} > \text{F} > \text{Br} > \text{I}$

Q55. Given below are two statements:

Statement I: $[\text{Fe}(\text{ox})_3]^{3-}$ is chiral.

Statement II: $\text{trans-}[\text{Cr}(\text{H}_2\text{O})_2(\text{ox})_2]^-$ is chiral.

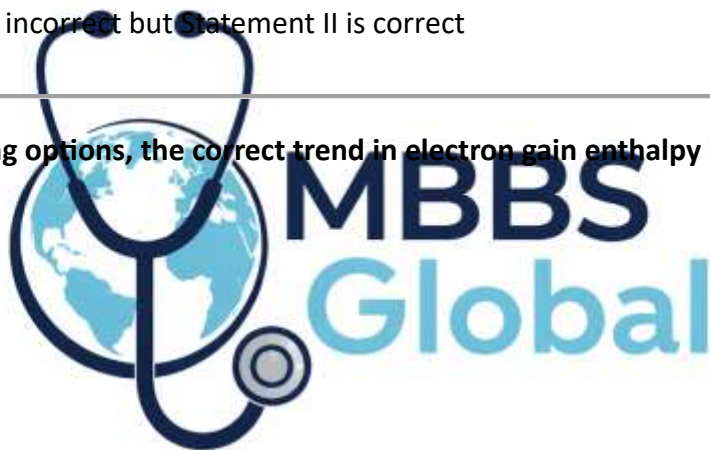
Choose the correct answer.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: D. Statement I is correct but Statement II is incorrect

Q56. The correct statement about peptides and proteins is:



Options:

- A. In α -helices, the polypeptide chain is twisted into a left-handed helix.
- B. Tertiary structure of proteins has two or more polypeptide subunits.
- C. Only proteins having quaternary structure are biologically active.
- D. In β -pleated sheet structures, peptide chains are held together by intermolecular hydrogen bonds.

Answer: D. In β -pleated sheet structures, peptide chains are held together by intermolecular hydrogen bonds.

Q57. Given below are two statements:

Statement I: Oxidation of p-nitrotoluene with acidic KMnO_4 gives an acid stronger than benzoic acid.

Statement II: Reduction of p-nitrotoluene with Sn/HCl followed by neutralization gives an amine more basic than aniline.

Choose the correct answer.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: B. Both Statement I and Statement II are correct

Q58. Identify the reactions which give aniline as the major product.

Options:

- A. A and B only
- B. B and D only
- C. A and C only
- D. C and D only

Answer: B. B and D only

Q59. Two moles of an ideal gas undergo free expansion from 10 L to 100 L at 300 K. The values of ΔS_{system} and $\Delta S_{\text{surroundings}}$ are:

Options:

- A. $\Delta S_{\text{system}} = 4.606R$; $\Delta S_{\text{surroundings}} = 0$
- B. $\Delta S_{\text{system}} = 0$; $\Delta S_{\text{surroundings}} = 0$
- C. $\Delta S_{\text{system}} = -4.606R$; $\Delta S_{\text{surroundings}} = 4.606R$
- D. $\Delta S_{\text{system}} = 0$; $\Delta S_{\text{surroundings}} = 4.606R$

Answer: A. $\Delta S_{\text{system}} = 4.606R$; $\Delta S_{\text{surroundings}} = 0$

Q60. The compound that CANNOT be obtained from the aldol condensation reaction shown is:

Options:

- A. Compound (1)
- B. Compound (2)
- C. Compound (3)
- D. Compound (4)

Answer: B. Compound (2)

Chemistry (Paper Code 70)

Q61. The complex which has facial and meridional isomers is:

(Given: py = pyridine and en = $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$)

Options:

- A. $[\text{Ni}(\text{en})_2(\text{H}_2\text{O})_2]^{2+}$
- B. $[\text{Cr}(\text{py})_3\text{Cl}_3]$
- C. $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
- D. $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{3+}$

Answer: B. $[\text{Cr}(\text{py})_3\text{Cl}_3]$

Q62. The numbers 17.0145 and 21.0235 rounded off to three figures after the decimal point are:

Options:

- A. 17.015 and 21.024
- B. 17.014 and 21.023
- C. 17.015 and 21.023
- D. 17.014 and 21.024

Answer: D. 17.014 and 21.024

Q63. The amount of carbon dioxide evolved upon complete combustion of 116 g of n-butane is:

(Given: H = 1, C = 12, O = 16)

Options:

- A. 362 g
- B. 352 g
- C. 322 g
- D. 176 g

Answer: B. 352 g

Q64. Consider the schematic plots of orbital wave function $\psi(r)$ against distance r from the nucleus. The figure representing two radial nodes is:



Options:

- A. D
- B. A
- C. B
- D. C

Answer: D. C

Q65. The following carbocation is stabilized by the interaction of the empty p orbital with:

Options:

- A. Empty σ^* and empty π^* orbitals
- B. Filled σ and filled π orbitals
- C. Empty σ and empty π^* orbitals
- D. Empty σ^* and filled π orbitals

Answer: B. Filled σ and filled π orbitals

Q66. A 1 : 3 electrolyte in aqueous solution is:

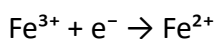
Options:

- A. $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$
- B. $[\text{CoCl}_3(\text{NH}_3)_4]\text{Cl}$
- C. $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
- D. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

Answer: D. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$



Q67. The standard electrode potential (E°) for the half-cell reaction:



at 298 K is:

(Given: $E^\circ(\text{Fe}^{3+}/\text{Fe}) = -0.04 \text{ V}$ and $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$)

Options:

- A. +0.92 V
- B. +0.40 V
- C. +0.76 V
- D. -0.48 V

Answer: C. +0.76 V

Q68. In potash alum, the ratio of K^+ and SO_4^{2-} ions is:

Options:

- A. 3 : 2

- B. 1 : 2
- C. 2 : 1
- D. 2 : 3

Answer: B. 1 : 2

Q69. Consider the following statements about solutions formed by mixing two liquids:

- A. An ideal solution obeys Raoult's law throughout the composition range.
- B. Mixture of chloroform and acetone shows negative deviation from Raoult's law.
- C. Mixture of aniline and phenol shows positive deviation from Raoult's law.

Options:

- A. A and C only
- B. A and B only
- C. B and C only
- D. A only

Answer: B. A and B only

Q70. For a salt XY (strong electrolyte), the plot of Λ_m versus $\sqrt{c}$ has a slope of $90.0 \text{ S cm}^2 \text{ mol}^{-1} \text{ L}^{1/2}$. At 0.01 M concentration, $\Lambda_m = 145.0 \text{ S cm}^2 \text{ mol}^{-1}$. The limiting molar conductivity of Y^- ion is:

(Given: $\lambda^\circ \text{X}^+ = 74.0 \text{ S cm}^2 \text{ mol}^{-1}$)

Options:

- A. 76.0
- B. 80.0
- C. 100.0
- D. 90.0

Answer: B. 80.0

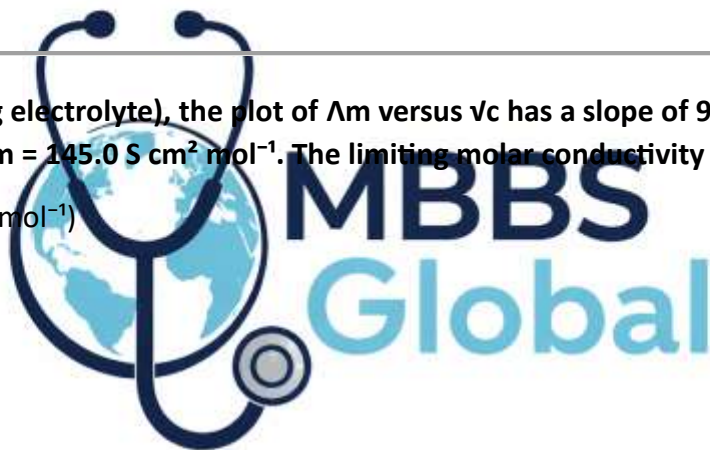
Q71. Arrange the following compounds in increasing order of polarity:

- A. $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$
- B. $\text{CH}_3\text{CH}_2\text{OH}$
- C. CH_3COCH_3
- D. CH_3COOH

Options:

- A. $\text{A} < \text{C} < \text{B} < \text{D}$
- B. $\text{A} < \text{B} < \text{C} < \text{D}$
- C. $\text{C} < \text{A} < \text{D} < \text{B}$
- D. $\text{C} < \text{A} < \text{B} < \text{D}$

Answer: A. $\text{A} < \text{C} < \text{B} < \text{D}$



Q72. According to Crystal Field Theory, the decreasing order of field strength of ligands is:

Options:

- A. $\text{Cl}^- > \text{NH}_3 > \text{H}_2\text{O} > \text{CO}$
- B. $\text{CO} > \text{NH}_3 > \text{H}_2\text{O} > \text{Cl}^-$
- C. $\text{CO} > \text{H}_2\text{O} > \text{NH}_3 > \text{Cl}^-$
- D. $\text{Cl}^- > \text{H}_2\text{O} > \text{NH}_3 > \text{CO}$

Answer: B. $\text{CO} > \text{NH}_3 > \text{H}_2\text{O} > \text{Cl}^-$

Q73. The amino acid that gives a blood-red colour with sodium nitroprusside test is:

Options:

- A. Serine
- B. Leucine
- C. Threonine
- D. Methionine

Answer: D. Methionine

Q74. In acidic medium, 10 mL of 0.25 M oxalic acid is titrated with KMnO_4 solution. If 10 mL of KMnO_4 is required, the strength of KMnO_4 solution is:

Options:

- A. 0.15 M
- B. 0.10 M
- C. 0.20 M
- D. 0.25 M

Answer: B. 0.10 M



Q75. The correct statement is:

Options:

- A. Aluminium has five valence orbitals.
- B. Boron has a maximum covalency of four.
- C. Beryllium has three valence orbitals.
- D. Magnesium has a maximum covalency of four.

Answer: B. Boron has a maximum covalency of four.

Chemistry (Paper Code 70)

Q76. Among the following, the compound having conjugated double bonds is:

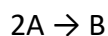
Options:

- A. Hepta-1,6-diene
- B. Hepta-1,3-diene

- C. Hepta-1,4-diene
D. Hepta-1,5-diene

Answer: B. Hepta-1,3-diene

Q77. For the zero-order reaction:



where $k = 1.0 \text{ mol L}^{-1} \text{ min}^{-1}$ and initial concentration of A is 2 M, the time required for 75% completion of the reaction is:

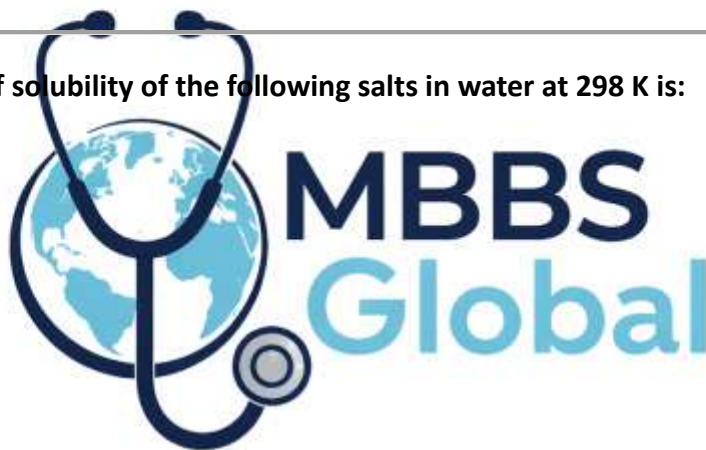
Options:

- A. 2.0 min
B. 1.5 min
C. 0.75 min
D. 1.0 min

Answer: C. 0.75 min

Q78. The correct order of solubility of the following salts in water at 298 K is:

Salt	K _{sp}
AgBr	5.0×10^{-13}
Zn(OH) ₂	1.0×10^{-15}
Hg ₂ Cl ₂	1.3×10^{-18}



Options:

- A. $\text{Zn(OH)}_2 > \text{AgBr} > \text{Hg}_2\text{Cl}_2$
B. $\text{Hg}_2\text{Cl}_2 > \text{Zn(OH)}_2 > \text{AgBr}$
C. $\text{AgBr} > \text{Zn(OH)}_2 > \text{Hg}_2\text{Cl}_2$
D. $\text{Hg}_2\text{Cl}_2 > \text{AgBr} > \text{Zn(OH)}_2$

Answer: A. $\text{Zn(OH)}_2 > \text{AgBr} > \text{Hg}_2\text{Cl}_2$

Q79. The correct decreasing order of oxidation state of the underlined atom is:

Options:

- A. $\text{P}_4\text{O}_6 > \text{Cl}_2\text{O}_7 > \text{AlH}_3$
B. $\text{P}_4\text{O}_{10} > \text{SO}_3 > \text{H}_2\text{O}$
C. $\text{N}_2\text{O}_5 > \text{Al}_2\text{O}_3 > \text{H}_2\text{S}$
D. $\text{PbO} > \text{N}_2\text{O}_3 > \text{SO}_2$

Answer: C. $\text{N}_2\text{O}_5 > \text{Al}_2\text{O}_3 > \text{H}_2\text{S}$

Q80. Consider the reversible processes for 1 mole of an ideal gas shown in the P–V diagram. If w_1 , w_2 , w_3 and w_4 represent the work done in processes 1, 2, 3 and 4 respectively, then the correct option is:

Options:

- A. $w_1 + w_2 + w_3 + w_4 = 0$
- B. $w_1 + w_3 = -2T_1 \ln(V_2/V_1) - 2T_2 \ln(V_4/V_3)$
- C. $w_2 + w_4 = \Delta U_2 - \Delta U_4$
- D. $w_1 + w_2 = 2T_1 \ln(V_2/V_1)$

Answer: B. $w_1 + w_3 = -2T_1 \ln(V_2/V_1) - 2T_2 \ln(V_4/V_3)$

Q81. Assertion (A): For an ideal solution formed by mixing liquids P and Q, $\Delta H_{\text{mix}} = 0$ and $\Delta V_{\text{mix}} = 0$.

Reason (R): No interactions occur between P and Q.

Options:

- A. A is not correct but R is correct
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are correct but R is NOT the correct explanation of A
- D. A is correct but R is not correct

Answer: D. A is correct but R is not correct

Q82. Among the following species, the spin-only magnetic moment is highest for:

(Given: Ti = 22, Mn = 25, Fe = 26, Co = 27)

Options:

- A. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$
- B. $[\text{Mn}(\text{CN})_6]^{3-}$
- C. $[\text{Fe}(\text{CN})_6]^{3-}$
- D. $[\text{Co}(\text{NH}_3)_6]^{3+}$

Answer: B. $[\text{Mn}(\text{CN})_6]^{3-}$

Q83. A protein undergoes reversible thermal denaturation:



At 60°C, concentrations of N and D are equal at equilibrium. If $\Delta H^\circ = 666 \text{ kJ mol}^{-1}$, then ΔS° is closest to:

Options:

- A. $11.1 \text{ kJ K}^{-1} \text{ mol}^{-1}$
- B. $2.0 \text{ kJ K}^{-1} \text{ mol}^{-1}$
- C. $2000.0 \text{ kJ K}^{-1} \text{ mol}^{-1}$
- D. $333.0 \text{ kJ K}^{-1} \text{ mol}^{-1}$

Answer: B. $2.0 \text{ kJ K}^{-1} \text{ mol}^{-1}$

Q84. Assertion (A): Generally, 3d transition metals have high melting points.

Reason (R): 3d electrons along with 4s electrons participate in metallic bonding.

Options:

- A. A is not correct but R is correct
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are correct but R is NOT the correct explanation of A
- D. A is correct but R is not correct

Answer: B. Both A and R are correct and R is the correct explanation of A

Q85. Assertion (A): The first ionization enthalpy of O is lower than that of N and F.

Reason (R): Removal of an electron from O leads to a stable half-filled p-orbital.

Options:

- A. A is not correct but R is correct
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are correct but R is NOT the correct explanation of A
- D. A is correct but R is not correct

Answer: B. Both A and R are correct and R is the correct explanation of A

Q86. Consider the reaction sequence involving:

- HBr / benzoyl peroxide
- Na / liquid NH_3
- H_2 / Pd-C (Lindlar catalyst)

Choose the correct statement.

Options:

- A. M and N are stereoisomers
- B. K and L are geometrical isomers
- C. K and L are enantiomers
- D. M and N are geometrical isomers

Answer: B. K and L are geometrical isomers

Q87. The highest occupied molecular orbital (HOMO) for Ne_2 is:

Options:

- A. σ_{2p}
- B. π_{2p}
- C. σ_{2p}^*
- D. π_{2p}^*

Answer: A. σ_{2p}

Q88. Match the species in List-I with their geometry in List-II.

List-I

- A. PCl_5
- B. BrF_5
- C. BF_4^-
- D. $[\text{Ni}(\text{CN})_4]^{2-}$

List-II

- I. Tetrahedral
- II. Square Planar
- III. Trigonal Bipyramidal
- IV. Square Pyramidal

Options:

- A. A–III, B–II, C–I, D–IV
- B. A–IV, B–III, C–I, D–II
- C. A–III, B–IV, C–I, D–II
- D. A–III, B–I, C–II, D–IV

Answer: C. A–III, B–IV, C–I, D–II



Q89. Match the vitamin with its source.

List-I

- A. Vitamin A
- B. Vitamin B_{12}
- C. Vitamin E
- D. Vitamin K

List-II

- I. Meat
- II. Sunflower oil
- III. Green leafy vegetables
- IV. Carrots

Options:

- A. A–III, B–I, C–IV, D–II
- B. A–II, B–III, C–IV, D–I
- C. A–IV, B–I, C–II, D–III
- D. A–IV, B–II, C–I, D–III

Answer: C. A–IV, B–I, C–II, D–III

Q90. For the following reaction sequence:

i. CH_3COCl / AlCl_3

ii. NaOCl

P + Q

Choose the correct statement.

Options:

A. Both P and Q are carbonyl compounds.

B. If P is the sodium salt of a carboxylic acid, Q is a primary alcohol.

C. P and Q are aromatic compounds.

D. If P gives a carboxylic acid on acidification, Q gives a poisonous gas on exposure to air and light.

Answer: D. If P gives a carboxylic acid on acidification, Q gives a poisonous gas on exposure to air and light.

End of Chemistry (Paper Code 70)



Biology (Paper Code 70)

Q91. Given below are two statements:

Statement I: The class name Reptilia refers to creeping or crawling mode of locomotion.

Statement II: All organisms belonging to Reptilia have three chambered heart.

Options:

A. Statement I is incorrect but Statement II is correct

B. Both Statement I and Statement II are correct

C. Both Statement I and Statement II are incorrect

D. Statement I is correct but Statement II is incorrect

Answer: D. Statement I is correct but Statement II is incorrect

Q92. How many turns of Calvin cycle are required for the formation of three molecules of glucose?

Options:

- A. 18
- B. 6
- C. 3
- D. 1

Answer: D. 1

Q93. Photorespiration reaction catalyzed by RuBisCo is shown below:



Identify "X" from the given options.

Options:

- A. Malate
- B. Phosphoenolpyruvate
- C. 2-Phosphoglycolate
- D. Oxaloacetate

Answer: C. 2-Phosphoglycolate



Q94. Given below are two statements:

Statement I: In gymnosperms, the male and female gametophytes remain within the sporangia.

Statement II: In gymnosperms, seeds are not covered.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: B. Both Statement I and Statement II are correct

Q95. How many molecules of pyruvic acid are produced at the end of glycolysis from 206 molecules of glucose?

Options:

- A. 412
- B. 206
- C. 309
- D. 103

Answer: A. 412

Q96. Match List-I with List-II.

List-I

A. Fusion of protoplasts between gametes I. Meiosis

B. Fusion of two nuclei

C. Generation of haploid spores

List-II

II. Plasmogamy

III. Karyogamy

Options:

A. A-I, B-III, C-II

B. A-II, B-III, C-I

C. A-II, B-I, C-III

D. A-III, B-II, C-I

Answer: B. A-II, B-III, C-I

Q97. Mitochondrial inner membrane encloses ____.

Options:

A. Aqueous humor

B. Matrix

C. Cytosol

D. Mucus

Answer: B. Matrix



Q98. Phyllotaxy is the pattern of arrangement of ____.

Options:

A. Sepals

B. Leaves

C. Flowers

D. Fruits

Answer: B. Leaves

Q99. Mad cow disease is caused by ____.

Options:

A. Mycoplasma sp.

B. Prions

C. Viroids

D. Aspergillus sp.

Answer: B. Prions

Q100. Cell theory was formulated by ____.

Options:

- A. Antonie Von Leeuwenhoek
- B. Schleiden and Schwann
- C. Robert Brown
- D. Singer and Nicolson

Answer: B. Schleiden and Schwann

Q101. Which of the following plant growth regulators promotes internode elongation prior to flowering in cabbage?

Options:

- A. Ethephon
- B. Absciscic acid
- C. Gibberellin
- D. Indole butyric acid

Answer: C. Gibberellin

Q102. Which pigment has absorption peak at 700 nm in the photosynthetic reaction centre PS I (P700)?

Options:

- A. Carotenoids
- B. Chlorophyll b
- C. Chlorophyll a
- D. Xanthophylls

Answer: C. Chlorophyll a

Q103. Sphenopsida class belongs to ____.

Options:

- A. Pteridophytes
- B. Bryophytes
- C. Angiosperms
- D. Gymnosperms

Answer: A. Pteridophytes

Q104. Which of the following represents the correct sequence of arrangement of bones in the lower limb of humans?

Options:

- A. Femur – Tarsal – Patella – Tibia



- B. Femur – Tibia – Patella – Tarsal
- C. Patella – Femur – Tibia – Tarsal
- D. Femur – Patella – Tibia – Tarsal

Answer: D. Femur – Patella – Tibia – Tarsal

Q105. Which of the following plant growth regulators is used as herbicide?

Options:

- A. Gibberellin
- B. 2,4-D
- C. Kinetin
- D. Absciscic acid

Answer: B. 2,4-D

Q106. Genus represents ____.

Options:

- A. A group of closely related families
- B. An individual plant or animal
- C. A population of plants and animals
- D. A group of closely related species

Answer: D. A group of closely related species



Q107. The plastid that stores xanthophyll is known as ____.

Options:

- A. Amyloplast
- B. Chloroplast
- C. Chromoplast
- D. Aleuroplast

Answer: C. Chromoplast

Q108. In water, frogs respire using ____.

Options:

- A. Trachea
- B. Skin
- C. Buccal cavity
- D. Lungs

Answer: B. Skin

Q109. Which of the following is not a characteristic of chordates?

Options:

- A. Presence of post anal part (tail)
- B. Presence of notochord
- C. Central nervous system is dorsal
- D. Absence of gills

Answer: D. Absence of gills

Q110. Smooth endoplasmic reticulum ____.

Options:

- A. Is a site for the synthesis of carbohydrates
- B. Has ribosomes attached to its surface
- C. Is the major site for the synthesis of lipids
- D. Is actively involved in protein synthesis

Answer: C. Is the major site for the synthesis of lipids

Q111. Which of the following are characteristics of prokaryotic cells?

- (a) Ribosomes are made of 50S and 30S subunits
- (b) They can have plasmids
- (c) They contain mesosome
- (d) They have peroxisomes

Options:

- A. (a), (b) and (c) only
- B. (b) and (c) only
- C. (a) and (c) only
- D. (a), (c) and (d) only

Answer: A. (a), (b) and (c) only

Q112. Match List-I with List-II.

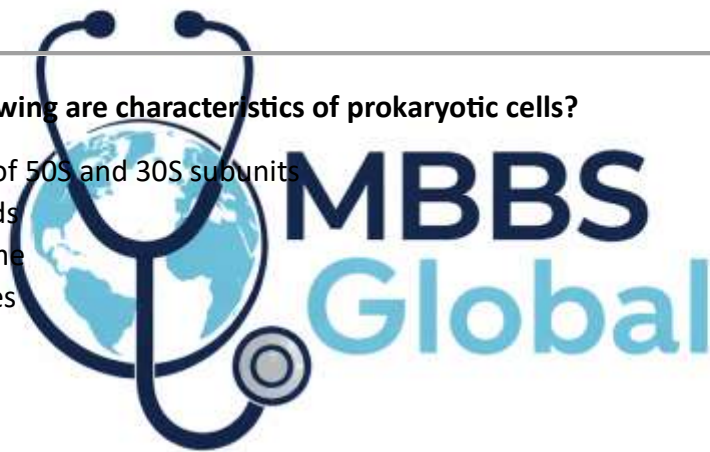
List-I

List-II

- | | |
|-----------------|--|
| A. Cristae | I. Flat membrane sacs in stroma of chloroplast |
| B. Cisternae | II. Infoldings in mitochondria |
| C. Thylakoids | III. Cell membrane |
| D. Phospholipid | IV. Disc shaped sacs in the Golgi apparatus |

Options:

- A. A-IV, B-III, C-I, D-II



- B. A-III, B-IV, C-I, D-II
- C. A-II, B-IV, C-I, D-III
- D. A-II, B-IV, C-III, D-I

Answer: C. A-II, B-IV, C-I, D-III

Q113. Which of the following statements related to pituitary gland are correct?

- (a) It is divided anatomically into adenohypophysis and neurohypophysis
- (b) It secretes follicle stimulating hormone
- (c) It secretes melanocyte stimulating hormone
- (d) It does not secrete prolactin

Options:

- A. (b) and (c) only
- B. (a) and (b) only
- C. (a), (b) and (c) only
- D. (c) and (d) only

Answer: C. (a), (b) and (c) only

Q114. Which of the following statements regarding photorespiration are correct?

- (a) Do not occur in C_3 plants
- (b) CO_2 is consumed and O_2 is generated
- (c) Phosphoglycolate is formed
- (d) No synthesis of ATP and NADPH

Options:

- A. (a) and (b) only
- B. (a) and (d) only
- C. (c) and (d) only
- D. (b) and (d) only

Answer: C. (c) and (d) only

Q115. Which of the following statements is incorrect?

Options:

- A. Fibrinogen is produced from fibrin
- B. Blood coagulates in response to an injury
- C. Blood clot consists of fibrins
- D. Fibrin is produced from fibrinogen

Answer: A. Fibrinogen is produced from fibrin

Q116. Arrange the following taxonomic categories in ascending order.



- (a) Genus
- (b) Class
- (c) Order
- (d) Phylum
- (e) Family
- (f) Kingdom
- (g) Species

Options:

- A. (f), (c), (b), (g), (d), (e), (a)
- B. (g), (a), (e), (c), (b), (d), (f)
- C. (a), (c), (d), (g), (f), (b), (e)
- D. (g), (c), (d), (b), (e), (a), (f)

Answer: B. (g), (a), (e), (c), (b), (d), (f)

Q117. Select the correct sequence of experiments that led to a gradual understanding of photosynthesis in green plants.

Options:

- A. Production of glucose → Role of air → Release of oxygen → Absorption spectra of chlorophyll a and b
- B. Absorption spectra of chlorophyll a and b → Production of glucose → Release of oxygen → Role of air
- C. Role of air → Release of oxygen → Production of glucose → Absorption spectra of chlorophyll a and b
- D. Release of oxygen → Production of glucose → Absorption spectra of chlorophyll a and b → Role of air

Answer: C. Role of air → Release of oxygen → Production of glucose → Absorption spectra of chlorophyll a and b

Q118. Match List-I with List-II.

List-I	List-II
A. Starch	I. Fights infection
B. Antibody	II. Energy storage
C. Concanavalin A	III. Glucose transport
D. Glut-4	IV. Lectin

Options:

- A. A-I, B-II, C-III, D-IV
- B. A-I, B-II, C-IV, D-III
- C. A-II, B-I, C-IV, D-III
- D. A-II, B-I, C-III, D-IV

Answer: C. A-II, B-I, C-IV, D-III

Q119. The number of vertebrae in a human is ____.

Options:

- A. 206
- B. 7
- C. 12
- D. 26

Answer: D. 26

Q120. Endomembrane system includes ____.

Options:

- A. Golgi complex, chloroplast, peroxisomes and vacuole
- B. Endoplasmic reticulum, Golgi complex, lysosomes and vacuole
- C. Endoplasmic reticulum, chloroplast, peroxisomes and vacuole
- D. Mitochondria, chloroplast, peroxisomes and vacuole

Answer: B. Endoplasmic reticulum, Golgi complex, lysosomes and vacuole

Biology (Paper Code 70)

Q121. Length of the stem at time 0 is 20 cm. The arithmetic growth rate is 30 cm per day. What is the length of the stem at the end of the 7th day?

Options:

- A. 460 cm
- B. 50 cm
- C. 170 cm
- D. 230 cm

Answer: D. 230 cm

Q122. Match List-I with List-II.

- | List-I | List-II |
|--------------|---------------|
| A. Spherical | I. Vibrio |
| B. Rod | II. Cocci |
| C. Comma | III. Spirilla |
| D. Spirillum | IV. Bacilli |

Options:

- A. A-II, B-IV, C-I, D-III
- B. A-I, B-III, C-II, D-IV



- C. A-III, B-II, C-I, D-IV
D. A-II, B-I, C-IV, D-III

Answer: A. A-II, B-IV, C-I, D-III

Q123. The number of action potentials generated by sino-arterial node (SAN) in a healthy human is _____ per minute.

Options:

- A. 120–140
B. 28–30
C. 70–75
D. 100–110

Answer: C. 70–75

Q124. Match List-I with List-II.

- | List-I | List-II |
|-----------|--------------------|
| A. Family | I. Sapindales |
| B. Genus | II. Dicotyledonae |
| C. Class | III. Anacardiaceae |
| D. Phylum | IV. Angiospermae |
| E. Order | V. Mangifera |



Options:

- A. A-III, B-V, C-II, D-IV, E-I
B. A-I, B-V, C-II, D-IV, E-III
C. A-II, B-I, C-III, D-IV, E-V
D. A-II, B-III, C-V, D-I, E-IV

Answer: A. A-III, B-V, C-II, D-IV, E-I

Q125. Which of the following is not a part of human central neural system?

Options:

- A. Pericardium
B. Arachnoid
C. Dura mater
D. Pia mater

Answer: A. Pericardium

Q126. Given below are two statements:

Statement I: Chromosomes are fully condensed at the end of prophase I.

Statement II: Meiosis I resembles mitosis.

Options:

- A. Statement I is incorrect, but Statement II is true
- B. Both Statement I and Statement II are true
- C. Both Statement I and Statement II are false
- D. Statement I is correct, but Statement II is false

Answer: D. Statement I is correct, but Statement II is false

Q127. Match List-I with List-II.

List-I

List-II

- | | |
|------------------------------|---------------|
| A. Marginal placentation | I. Argemone |
| B. Axile placentation | II. Tomato |
| C. Parietal placentation | III. Primrose |
| D. Free central placentation | IV. Pea |

Options:

- A. A-IV, B-II, C-I, D-III
- B. A-II, B-IV, C-I, D-III
- C. A-IV, B-II, C-III, D-I
- D. A-IV, B-III, C-I, D-II

Answer: A. A-IV, B-II, C-I, D-III



Q128. Symbiotic association between fungi and algae are called ____.

Options:

- A. Chrysophytes
- B. Lichens
- C. Sponges
- D. Mycorrhiza

Answer: B. Lichens

Q129. Which of the following is not a prokaryote?

Options:

- A. Fungi
- B. Bacteria

- C. Blue green algae
- D. Mycoplasma

Answer: A. Fungi

Q130. Arrange the following elements in descending order of their contribution to percentage weight of the human body.

- (a) Oxygen
- (b) Carbon
- (c) Hydrogen
- (d) Nitrogen

Options:

- A. (b), (a), (c), (d)
- B. (a), (b), (c), (d)
- C. (c), (a), (b), (d)
- D. (b), (c), (d), (a)

Answer: B. (a), (b), (c), (d)

Q131. Which one of the following statements is incorrect?

Options:

- A. β -cells of pancreas secrete insulin
- B. α -cells of pancreas secrete glucagon
- C. α -cells of pancreas secrete insulin
- D. Glucagon stimulates glycogenolysis

Answer: C. α -cells of pancreas secrete insulin

Q132. Which of the following are characteristic features of Solanaceae family?

- (a) Flowers are bisexual and actinomorphic
- (b) Calyx have five sepals and are united
- (c) Androecium have five stamens and are epipetalous
- (d) Ovary is inferior

Options:

- A. (b), (c) and (d) only
- B. (a), (b) and (c) only
- C. (d) only
- D. (a) and (b) only

Answer: B. (a), (b) and (c) only

Q133. Given below are two statements:



Statement I: When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry.

Statement II: In phylum Echinodermata, both adults and larvae are radially symmetrical.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: D. Statement I is correct but Statement II is incorrect

Q134. The correct sequence of adult cell cycle phases is ____.

Options:

- A. S-M-G₂-G₁
- B. G₁-G₂-S-M
- C. G₁-M-G₂-S
- D. G₁-S-G₂-M

Answer: D. G₁-S-G₂-M

Q135. In frogs, the number of pairs of cranial nerves arising from the brain are ____.

Options:

- A. 12
- B. 6
- C. 9
- D. 10

Answer: D. 10

Q136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: In recombinant DNA technology, lysozyme is used for disrupting bacterial cells while cellulase is for plant cells.

Reason R: Isolation of genetic material needs disruption of cells.

Options:

- A. A is not correct but R is correct
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are correct but R is not the correct explanation of A
- D. A is correct but R is not correct

Answer: B. Both A and R are correct and R is the correct explanation of A



Q137. The method of directly injecting a sperm into ovum in assisted reproductive technology is called:

Options:

- A. Embryo Transfer (ET)
- B. Gamete Intra Fallopian Transfer (GIFT)
- C. Zygote Intra Fallopian Transfer (ZIFT)
- D. Intra Cytoplasmic Sperm Injection (ICSI)

Answer: D. Intra Cytoplasmic Sperm Injection (ICSI)

Q138. Adaptive radiation in placental mammals and Australian marsupials leading to similarity between distant species is an example of ____.

Options:

- A. Genetic drift
- B. Divergent evolution
- C. Convergent evolution
- D. Founder effect

Answer: C. Convergent evolution

Q139. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: In an experiment, Mendel observed that the F_1 progeny plants are all tall and none are dwarf.

Reason R: Stem height is a contrasting trait, with tall being dominant and dwarf being recessive.

Options:

- A. A is not correct but R is correct
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are correct but R is not the correct explanation of A
- D. A is correct but R is not correct

Answer: B. Both A and R are correct and R is the correct explanation of A

Q140. Arrange the following in descending order of number of species in the Amazonian rain forest.

- (a) Plants
- (b) Birds
- (c) Fishes
- (d) Invertebrates
- (e) Mammals

Options:

- A. (b) > (a) > (d) > (c) > (e)
- B. (c) > (b) > (d) > (e) > (a)
- C. (d) > (a) > (c) > (b) > (e)
- D. (e) > (b) > (a) > (c) > (d)

Answer: C. (d) > (a) > (c) > (b) > (e)

Q141. Sponges exchange O₂ with CO₂ by _____.

Options:

- A. Gills
- B. Simple diffusion over their entire body surfaces
- C. Moist cuticle
- D. Tracheal tubes

Answer: B. Simple diffusion over their entire body surfaces

Q142. For a person with blood group 'O', which of the following is not a possible combination of parent's blood group genotypes?

Options:

- A. Father: IAIB and Mother: IAi
- B. Father: IAi and Mother: IBi
- C. Father: IAi and Mother: IAi
- D. Father: IBi and Mother: IBi

Answer: A. Father: IAIB and Mother: IAi



Q143. Given below are two statements:

Statement I: Modern Homo sapiens arose in Australia and moved across continents.

Statement II: Homo sapiens arose around 75000 to 10000 years ago.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: A. Statement I is incorrect but Statement II is correct

Q144. Which of the following is used as an effective sedative and painkiller for treating post-surgery patients?

Options:

- A. Anti-retroviral drugs
- B. Interferon
- C. Antibiotics
- D. Morphine

Answer: D. Morphine

Q145. Which of the following plant produces non-albuminous seeds?

Options:

- A. Pea
- B. Wheat
- C. Maize
- D. Barley

Answer: A. Pea

Q146. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Abingdon tortoise in Galapagos islands became extinct within a decade after goats were introduced.

Reason R: Goats were more efficient at browsing than Abingdon tortoise.

Options:

- A. A is not correct but R is correct
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are correct but R is not the correct explanation of A
- D. A is correct but R is not correct

Answer: B. Both A and R are correct and R is the correct explanation of A

Q147. The covering of ovum at ovulation is _____.

Options:

- A. Chorion
- B. Endometrium
- C. Zona radiata
- D. Zona pellucida

Answer: D. Zona pellucida

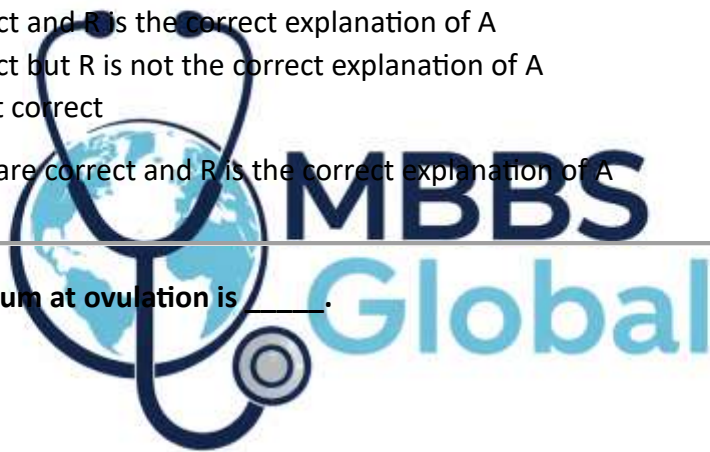
Q148. Which of the following is used as a clot buster?

Options:

- A. Statins
- B. Streptokinase
- C. Penicillin
- D. Cyclosporin A

Answer: B. Streptokinase

Q149. Which of the following structure is not a part of the male reproductive system?



Options:

- A. Infundibulum
- B. Rete testis
- C. Epididymis
- D. Vasa efferentia

Answer: A. Infundibulum

Q150. Given below are two statements:

Statement I: Ovulation is caused by LH surge leading to rupture of Graafian follicles.

Statement II: Graafian follicle remaining after ovulation transforms into corpus luteum and secretes large amount of estrogen.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: D. Statement I is correct but Statement II is incorrect

Biology (Paper Code 70)

Q151. The opening between the right atrium and the right ventricle is guarded by ____.

Options:

- A. Sino-atrial node
- B. Bicuspid valve
- C. Tricuspid valve
- D. Semilunar valve

Answer: C. Tricuspid valve

Q152. Which of the following is not evidence for evolution?

Options:

- A. Divergent evolution of anatomical structures such as forelimbs
- B. Convergent evolution of traits like wings of birds and butterflies
- C. Paleontological evidence from fossil records
- D. Embryological support for evolution as proposed by Ernst Haeckel

Answer: D. Embryological support for evolution as proposed by Ernst Haeckel

Q153. The inactive form of Bt toxin is converted to the active form in the insect gut ____.

Options:

- A. By nucleases

- B. Due to alkaline pH
- C. Due to acidic pH
- D. By proteases

Answer: B. Due to alkaline pH

Q154. Colostrum, secreted by mother during initial days of lactation, is abundant in ____.

Options:

- A. IgD
- B. IgG
- C. IgM
- D. IgA

Answer: D. IgA

Q155. Which of the following in female gametophyte of an angiosperm helps in guiding the pollen tube for fertilizing the egg?

Options:

- A. Polar nucleus
- B. Antipodals
- C. Synergids
- D. Central cell

Answer: C. Synergids



Q156. Which of the following disease is not sexually transmitted?

Options:

- A. Genital warts
- B. Syphilis
- C. Tuberculosis
- D. Gonorrhoea

Answer: C. Tuberculosis

Q157. Which of the following statements about lac-operon is correct?

Options:

- A. Galactose can act as an inducer of lac operon
- B. Gene i is constitutively expressed
- C. Lactose activates repressor to bind to the operator
- D. Genes i, z, y and a share single common promoter

Answer: B. Gene i is constitutively expressed

Q158. Match List-I with List-II.

List-I

List-II

- | | |
|-------------------|-----------------------------------|
| A. Transformation | I. Restriction enzyme |
| B. Cloning site | II. Transfer DNA to host bacteria |
| C. Selection | III. Replication |
| D. Ori | IV. Antibiotic |

Options:

- A. A-IV, B-I, C-III, D-II
- B. A-II, B-I, C-IV, D-III
- C. A-I, B-II, C-IV, D-III
- D. A-III, B-IV, C-II, D-I

Answer: B. A-II, B-I, C-IV, D-III

Q159. A population of diploid organisms is at Hardy-Weinberg equilibrium. If the frequency of allele A is 0.1, the frequency of AA is ____.

Options:

- A. 0.99
- B. 0.01
- C. 0.02
- D. 0.10

Answer: B. 0.01



Q160. Sperm motility is due to ____.

Options:

- A. Muscular movement
- B. Flagellar movement
- C. Ciliary movement
- D. Amoeboid movement

Answer: B. Flagellar movement

Q161. Consider a population of 10 million cells. Given the per-capita birth rate of 0.002 (per unit time) and the per-capita death rate of 0.002 (per unit time), the expected number of cells after 10 generations is ____.

Options:

- A. 100 million
- B. 1 million

- C. 5 million
- D. 10 million

Answer: D. 10 million

Q162. Given below are two statements: one is labelled as Assertion A and the other as Reason R.

Assertion A: Forelimbs of humans and bats are homologous.

Reason R: Forelimbs of humans and bats have similar anatomical structure.

Options:

- A. A is false but R is true
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are true, but R is not the correct explanation of A
- D. A is true but R is false

Answer: B. Both A and R are correct and R is the correct explanation of A

Q163. Muscle contraction is initiated by a signal sent by the central nervous system by the release of ____.

Options:

- A. Cyclic adenine monophosphate
- B. Acetylcholine
- C. Acetyl coenzyme A
- D. Cyclic guanine monophosphate

Answer: B. Acetylcholine



Q164. Which of the following hormone is not secreted by human placenta?

Options:

- A. LH
- B. hCG
- C. Estrogen
- D. Progesterone

Answer: A. LH

Q165. Which of the following statements is correct about *Plasmodium*?

Options:

- A. Fertilization takes place in mosquito gut
- B. Reproduces sexually in liver cells
- C. Reproduces sexually in RBCs
- D. Gametocytes develop in mosquito gut

Answer: A. Fertilization takes place in mosquito gut

Q166. Which of the following are primary consumers in a food chain?

Options:

- A. Carnivores
- B. Parasites
- C. Predators
- D. Herbivores

Answer: D. Herbivores

Q167. Which of the following statements about the reabsorption process in Henle's loop are correct?

- (a) The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes.
- (b) Urine gets concentrated in Henle's loop.
- (c) Reabsorption of Na^+ and water takes place in Henle's loop.
- (d) Active or passive transport of electrolytes occurs in the ascending limb of Henle's loop.

Options:

- A. (a), (b) and (d) only
- B. (a) and (b) only
- C. (b), (c) and (d) only
- D. (a), (b) and (c) only

Answer: A. (a), (b) and (d) only



Q168. Given below are two statements: one is labelled as Assertion A and the other as Reason R.

Assertion A: The logistic growth model of populations is considered more realistic than the exponential growth model.

Reason R: Resources are finite.

Options:

- A. A is not correct but R is correct
- B. Both A and R are correct and R is the correct explanation of A
- C. Both A and R are correct but R is not the correct explanation of A
- D. A is correct but R is not correct

Answer: B. Both A and R are correct and R is the correct explanation of A

Q169. Which of the following is the correct order of arrangement of vertebral column from head to toe?

Options:

- A. Cervical vertebra → Thoracic vertebra → Lumbar vertebra → Sacrum
- B. Cervical vertebra → Thoracic vertebra → Sacrum → Lumbar vertebra
- C. Sacrum → Lumbar vertebra → Thoracic vertebra → Cervical vertebra
- D. Cervical vertebra → Lumbar vertebra → Thoracic vertebra → Sacrum

Answer: A. Cervical vertebra → Thoracic vertebra → Lumbar vertebra → Sacrum

Q170. Match List-I with List-II.

List-I

- A. Both species are harmed
- B. One species is harmed and the other is benefited
- C. Both species are benefited
- D. One is benefited while the other has no effect

List-II

- I. Predation
- II. Mutualism
- III. Competition
- IV. Commensalism

Options:

- A. A-III, B-I, C-II, D-IV
- B. A-III, B-IV, C-II, D-I
- C. A-I, B-II, C-III, D-IV
- D. A-II, B-I, C-IV, D-III

Answer: A. A-III, B-I, C-II, D-IV

Q171. If the diploid chromosome number of a typical angiosperm is 36, what would be the chromosome number in its endosperm?

Options:

- A. 72
- B. 18
- C. 36
- D. 54

Answer: D. 54

Q172. Which of the following enzymes synthesizes precursor mRNA?

Options:

- A. DNA polymerase
- B. RNA polymerase I
- C. RNA polymerase II
- D. RNA polymerase III

Answer: C. RNA polymerase II

Q173. Given below are two statements:

Statement I: Plasmids are autonomously replicating DNA.

Statement II: Plasmids are extrachromosomal DNA.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: B. Both Statement I and Statement II are correct

Q174. How many theca are present in each lobe of a typical bilobed angiosperm anther?

Options:

- A. 12
- B. 2
- C. 6
- D. 8

Answer: B. 2

Q175. Natural selection can lead to ____.

- (a) Stabilisation
- (b) Genetic drift
- (c) Directional change
- (d) Disruption

Options:

- A. (a) and (c) only
- B. (a) only
- C. (a), (c) and (d) only
- D. (a), (b), (c) and (d)

Answer: C. (a), (c) and (d) only



Q176. Which of the following statements are correct?

- (a) Energy flow from producers to consumers is unidirectional
- (b) Energy pyramid can never be inverted
- (c) Transfer of energy follows the 1% law

Options:

- A. (b) and (c) only
- B. (a), (b) and (c)
- C. (a) and (b) only
- D. (a) and (c) only

Answer: C. (a) and (b) only

Q177. Match List-I with List-II.

List-I

List-II

- | | |
|--------------------------|---|
| A. Excess growth hormone | I. Reabsorption of water and electrolytes in kidney |
| B. Luteinizing hormone | II. Contraction of uterus during childbirth |
| C. Vasopressin | III. Acromegaly |
| D. Oxytocin | IV. Ovulation |

Options:

- A. A-IV, B-III, C-I, D-II
- B. A-III, B-IV, C-II, D-I
- C. A-III, B-IV, C-I, D-II
- D. A-II, B-IV, C-I, D-III

Answer: C. A-III, B-IV, C-I, D-II

Q178. Which of the following are secondary lymphoid organs?

- (a) Bone marrow
- (b) Tonsils
- (c) Spleen
- (d) Thymus

Options:

- A. (a) and (d) only
- B. (a) and (b) only
- C. (b) and (c) only
- D. (b) and (d) only

Answer: C. (b) and (c) only



Q179. During PCR, primers bind to the DNA strands in the _____ step.

Options:

- A. Ligation
- B. Denaturation
- C. Extension
- D. Annealing

Answer: D. Annealing

Q180. Given below are two statements:

Statement I: Down's syndrome is caused by the absence of one of the X-chromosomes.

Statement II: Turner's syndrome is caused by the presence of an additional copy of the chromosomes.

Options:

- A. Statement I is incorrect but Statement II is correct
- B. Both Statement I and Statement II are correct
- C. Both Statement I and Statement II are incorrect
- D. Statement I is correct but Statement II is incorrect

Answer: C. Both Statement I and Statement II are incorrect

